

Course Code: Ethical Hacking (ITA1404)

Software Required: Kali Linux

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Experiment 1

Basic File and User Management Commands

Aim: To execute and understand the usage of **rm**, **users**, and **tree** commands in Kali Linux.

Procedure:

1. Open Kali Linux Terminal.
2. Create a sample file.
3. Execute the rm command.
4. Execute the users command.
5. Execute the tree command.

Command:

1. touch test.txt
2. ls test.txt
3. rm test.txt
4. ls test.txt
5. users
6. tree

Output:

```
(kali@kali)-[~]
└─$ touch aswatha.txt

(kali@kali)-[~]
└─$ ls aswatha.txt
aswatha.txt

(kali@kali)-[~]
└─$ rm aswatha.txt

(kali@kali)-[~]
└─$ ls aswatha.txt
ls: cannot access 'aswatha.txt': No such file or directory

(kali@kali)-[~]
```

Result:

Successfully executed rm, users, and tree commands in Kali Linux.

Experiment 2

File Viewing Commands

Aim: To execute and understand the usage of **less**, **more** and **vi** commands.

Procedure:

1. Create a text file.
2. Open the file using less.
3. Open the file using more.
4. Edit the file using vi.

Command:

1. vi sample.txt
2. less sample.txt
3. more sample.txt
4. vi sample.txt

Output:

```
(kali㉿kali)-[~]
└─$ vi sample.txt
(kali㉿kali)-[~]
└─$ less sample.txt
(kali㉿kali)-[~]
└─$ more sample
more: cannot open sample: No such file or directory
(kali㉿kali)-[~]
└─$ more sample.txt
hello welcome
(kali㉿kali)-[~]
└─$ vi sample.txt
(kali㉿kali)-[~]
└─$ vi sample.txt
(kali㉿kali)-[~]
└─$ less sample.txt
(kali㉿kali)-[~]
└─$ more sample.txt
hello welcomeyeerwteyr e
(kali㉿kali)-[~]
└─$
```

Result:

Successfully executed file viewing and editing commands.

Experiment 3

Tree, Head and Tail Commands

Aim: To execute and understand the usage of tree, head, and tail commands.

Procedure:

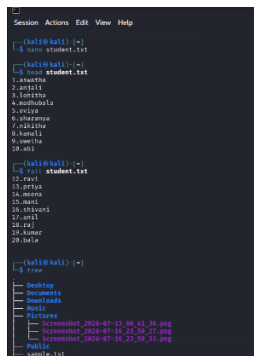
1. Create and open the file using:
nano student.txt
2. Type the content (names) inside the file.

3. Save the file:
Press Ctrl + O, then press Enter
4. Exit the editor:
Press Ctrl + X
5. Display first 10 lines:
head student.txt
6. Display last 10 lines:
tail student.txt
7. Display directory structure:
tree

Commands:

1. nano student.txt
2. head student.txt
3. tail student.txt
4. tree

Output:



```
kali@kali: ~$ nano student.txt
kali@kali: ~$ head student.txt
1.anna
2.anna
3.anna
4.anna
5.anna
6.anna
7.anna
8.anna
9.anna
10.anna
kali@kali: ~$ tail student.txt
12.anna
13.anna
14.anna
15.anna
16.anna
17.anna
18.anna
19.anna
20.anna
kali@kali: ~$ tree
tree
.
├── Desktop
├── Documents
├── Downloads
├── Music
├── Pictures
├── Screenshots
├── Templates
├── Public
└── user.txt
```

Result:

Successfully executed tree, head, and tail commands.

Experiment 4

Directory and File Display Commands

Aim: To execute ls, cat, and mkdir commands.

Procedure:

1. Create a directory (mkdir EthicalHacking)
2. Move into the directory (cd EthicalHacking)
3. Check directory contents (ls)
4. Create and open a file (nano file.txt)
5. Type the content inside the file
6. Save the file (Ctrl + O → Enter)

7. Exit nano editor (Ctrl + X)
8. Check file exists (ls)
9. Display file content (cat file.txt)

Code:

1. mkdir EthicalHacking
2. cd EthicalHacking
3. ls
4. nano file.txt
5. ls
6. cat file.txt

Output:

```
(kali@kali)-[~]
$ mkdir EthicalHacking
(kali@kali)-[~]
$ cd EthicalHacking
(kali@kali)-[~/EthicalHacking]
$ ls
(kali@kali)-[~/EthicalHacking]
$ nano file.txt
(kali@kali)-[~/EthicalHacking]
$ ls
file.txt
(kali@kali)-[~/EthicalHacking]
$ cat file.txt
this is my first file.
i am learning Kali Linux commands.
this is a CAT command experiment.
(kali@kali)-[~/EthicalHacking]
$
```

Result:

Successfully executed ls, cat, and mkdir commands.

Experiment 5

System Information Commands

Aim: To execute **uptime**, **date**, and **mv** commands.

Procedure:

1. Check system uptime (uptime)
2. Display current date and time (date)
3. Create a file (touch f1.txt)
4. Rename the file (mv f1.txt f2.txt)
5. Verify the file (ls)

Commands:

1. uptime
2. date
3. touch f1.txt
4. mv f1.txt f2.txt

5. ls

Output:

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ uptime
23:16:34 up 6 min, 1 user, load average: 1.34, 1.12, 0.59
kali@kali:~$ date
Tue Jul 21 11:16:37 PM EDT 2026
kali@kali:~$ touch f2.txt
kali@kali:~$ mv f1.txt f2.txt
kali@kali:~$ ls
Desktop  Documents  Downloads  EthicalHacking  f2.txt  Music  Pictures  Public  sample.txt  sample.txt  student.txt  Templates  Videos
kali@kali:~$
```

Result:

Successfully executed uptime, date, and mv commands.

Experiment 6

File Creation and Download Commands

Aim: To execute touch, pwd, and wget commands.

Procedure:

1. Create a new file:
touch sample.txt
2. Display current directory:
pwd
3. Download a file:
wget https://example.com/file.zip

Commands:

1. touch sample.txt
2. pwd
3. wget https://example.com/file.zip

Output:

```
Session Actions Edit View Help
(kali@kali)-[~]
└─$ touch sample.txt
(kali@kali)-[~]
└─$ ll
total 140
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Desktop
drwxr-xr-x 2 kali kali 4096 Jul 23 10:38 Documents
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Downloads
drwxrwxr-x 2 kali kali 4096 Jul 17 00:11 EthicalHacking
-rw-rw-r-- 1 kali kali 0 Jul 21 23:16 f2.txt
-rw-rw-r-- 1 kali kali 50 Jul 23 11:01 file.txt
-rw-rw-r-- 1 kali kali 83042 Jul 26 23:36 index.html
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Music
-rw-rw-r-- 1 kali kali 28 Jul 23 11:19 passwords.txt
drwxr-xr-x 2 kali kali 4096 Jul 21 23:14 Pictures
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Public
-rw-rw-r-- 1 kali kali 22 Jul 13 00:55 sample.tst
-rw-rw-r-- 1 kali kali 27 Jul 26 23:36 sample.txt
-rw-rw-r-- 1 kali kali 182 Jul 16 23:58 student.txt
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Templates
drwxr-xr-x 2 kali kali 4096 Jan 20 2026 Videos
(kali@kali)-[~]
└─$ pwd
/home/kali
(kali@kali)-[~]
└─$ wget https://www.google.com/index.html
--2026-07-26 23:36:35-- https://www.google.com/index.html
Resolving www.google.com (www.google.com)... 142.251.157.119, 142.251.151.119, 142.251.154.119, ...
Connecting to www.google.com (www.google.com)|142.251.157.119|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html.1'

index.html.1      [  =>
2026-07-26 23:36:35 (2.26 MB/s) - 'index.html.1' saved [83176]
```

Result:

Successfully executed touch, pwd, and wget commands.

Experiment 7

Navigation and Copy Commands

Aim: To execute cd, cp, and ls commands.

Procedure:

1. Change directory (cd Documents – moves into Documents folder)
2. Copy a file (cp file1.txt backup.txt – creates duplicate file)
3. List files (ls – displays files to verify copy)

Commands:

1. cd Documents
2. cp file1.txt backup.txt
3. ls

Output:

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ pwd  
/home/kali  
(kali@kali)-[~]  
$ ls  
Desktop Documents Downloads EthicalHacking f2.txt Music Pictures Public sample.txt sample.txt student.txt Templates Videos  
(kali@kali)-[~]  
$ cd Documents  
(kali@kali)-[~/Documents]  
$ ls  
(kali@kali)-[~/Documents]  
$ touch file1.txt  
(kali@kali)-[~/Documents]  
$ ls  
file1.txt  
(kali@kali)-[~/Documents]  
$ cp file1.txt backup.txt  
(kali@kali)-[~/Documents]  
$ ls  
backup.txt file1.txt  
(kali@kali)-[~/Documents]  
$ cd ..  
cd: string not in pwd: .  
(kali@kali)-[~/Documents]  
$ cd ..  
(kali@kali)-[~]  
$
```

Result:

Successfully executed cd, cp, and ls commands.

Experiment 8

Nmap Service Version and OS Detection

Aim: To perform service version detection and operating system identification using Nmap.

Procedure:

1. Detect service versions (nmap -sV 192.168.1.10 – shows service names and versions)
2. Perform aggressive scan (nmap -A 192.168.1.10 – includes OS, version, scripts, traceroute)
3. Detect operating system (sudo nmap -O 192.168.1.10 – identifies OS details)

Commands:

- 1) nmap -sV 192.168.1.10
- 2) nmap -A 192.168.1.10
- 3) sudo nmap -O 192.168.1.10

Output:

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)~]
$ nmap -sV 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 10:45 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000045s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.58 seconds

(kali@kali)~]
$ nmap -A 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 10:46 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00028s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.89 seconds
```

Result:

Successfully performed service version detection and OS identification using Nmap.

Experiment 9

Nmap Timing and Performance Commands

Aim: To perform IDS evasion and timing scans using Nmap.

Procedure:

- 1) Perform polite scan (nmap -T2 192.168.1.10 – slower and stealthy)
- 2) Perform normal scan (nmap -T3 192.168.1.10 – default balanced scan)

Commands:

- 1) nmap -T2 192.168.1.10
- 2) nmap -T3 192.168.1.10

Output:

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
└─$ nmap -T2 127.0.0.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 10:53 EDT  
Stats: 0:00:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 4.20% done; ETC: 11:01 (0:07:13 remaining)  
Stats: 0:00:21 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 4.80% done; ETC: 11:01 (0:07:16 remaining)  
Stats: 0:00:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 5.60% done; ETC: 11:01 (0:07:01 remaining)  
Stats: 0:00:26 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 5.90% done; ETC: 11:01 (0:07:11 remaining)  
Stats: 0:00:26 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 6.00% done; ETC: 11:01 (0:07:03 remaining)  
Stats: 0:00:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 6.70% done; ETC: 11:01 (0:07:12 remaining)  
Stats: 0:00:35 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 8.00% done; ETC: 11:01 (0:06:54 remaining)  
Stats: 0:00:36 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 8.20% done; ETC: 11:01 (0:06:54 remaining)  
Stats: 0:00:36 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 8.30% done; ETC: 11:01 (0:06:49 remaining)  
Stats: 0:00:37 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 8.40% done; ETC: 11:01 (0:06:54 remaining)  
Stats: 0:00:37 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 8.50% done; ETC: 11:01 (0:06:49 remaining)  
Stats: 0:00:38 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 8.60% done; ETC: 11:01 (0:06:54 remaining)  
Stats: 0:00:38 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 8.70% done; ETC: 11:01 (0:06:49 remaining)  
Stats: 0:00:38 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 8.80% done; ETC: 11:01 (0:06:44 remaining)  
  
(kali@kali)-[~]  
└─$ nmap -T3 127.0.0.1  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 10:54 EDT  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000050s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.45 seconds
```

Result:

Successfully executed Nmap timing and performance scans in Kali Linux.

Experiment 10: Free, Sort and History Commands

Aim: To execute and understand the usage of Free, Sort, and History commands in Kali Linux.

Procedure:

- 1) Display memory usage (free – shows RAM usage)
- 2) Display in readable format (free -h – human readable output)
- 3) Sort file (sort file.txt – arranges in ascending order)
- 4) Reverse sort (sort -r file.txt – descending order)
- 5) View history (history – shows executed commands)

Commands:

- 1) free
- 2) free -h
- 3) sort file.txt
- 4) sort -r file.txt
- 5) history

Output:

```
Session Actions Edit View Help
(kali@kali)-[~]
$ free
              total        used        free      shared  buff/cache   available
Mem:           2018416       928344       744912         20684        509704       1090072
Swap:           976556           0         976556

(kali@kali)-[~]
$ nano file.txt

(kali@kali)-[~]
$ sort file.txt

apple
banana
blueberry
cherry
mango
melons
peach

(kali@kali)-[~]
$ sort -r file.txt

peach
melons
mango
cherry
blueberry
banana
apple

(kali@kali)-[~]
$ history 10
1 cd /usr/share/wordlists/metasploit/
2 ls
3 ping 192.168.1.101
4 ls
5 touch class.txt
6 ls
7 nano class.txt
8 nmap -sV 127.0.0.1
9 nmap -A 127.0.0.1
10 sudo nmap -o 127.0.0.1
```

Result:

Successfully executed Free, Sort, and History commands in Kali Linux.

Experiment 11: Nmap Timing and Performance

Aim: To perform aggressive speed scanning using Nmap.

Procedure:

- 1) Perform aggressive scan (nmap -T4 192.168.1.1 – faster scanning)
- 2) Perform insane scan (nmap -T5 192.168.1.1 – very fast, less stealth)

Commands:

- 1) nmap -T4 192.168.1.1
- 2) nmap -T5 192.168.1.1

Output:

```
(kali㉿kali)-[~]
$ nmap -T4 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 11:04 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000080s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds

(kali㉿kali)-[~]
$ nmap -T5 127.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 11:04 EDT
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000070s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
```

Result:

Successfully executed Nmap insane speed scan.

Experiment 12: Port Scanning Tools

Aim: To identify open ports on a target system.

Procedure:

- 1) Perform basic scan (nmap 192.168.1.1 – lists open ports)
- 2) Perform SYN scan (nmap -sS 192.168.1.1 – stealth port scan)

Commands:

- 1) nmap 192.168.1.1
- 2) nmap -sS 192.168.1.1

Output:

```
Session Actions Edit View Help
kali@kali:~/Documents$ cat /dev/null > /dev/null
kali@kali:~/Documents$ nmap 12.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 11:09 EDT
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 31.35% done; ETC: 11:10 (0:00:07 remaining)
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 33.85% done; ETC: 11:10 (0:00:06 remaining)
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 54.90% done; ETC: 11:10 (0:00:03 remaining)
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 77.30% done; ETC: 11:10 (0:00:01 remaining)
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 89.75% done; ETC: 11:10 (0:00:01 remaining)
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 97.60% done; ETC: 11:10 (0:00:00 remaining)
Nmap scan report for 12.0.0.1
Host is up (0.0059s latency).
All 1000 scanned ports on 12.0.0.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 13.00 seconds

kali@kali:~/Documents$ nmap -sS 12.0.0.1
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 11:10 EDT
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:03 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:05 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 85.75% done; ETC: 11:10 (0:00:01 remaining)
Nmap scan report for 12.0.0.1
Host is up (0.0027s latency).
All 1000 scanned ports on 12.0.0.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 11.72 seconds
```

Result:

Open ports and running services were identified.

Experiment 13: Host Discovery

Aim: To discover active hosts in a network.

Procedure:

- 1) Discover active hosts (nmap -sn 192.168.1.0/24 – identifies live systems in network).

Commands:

- 1) nmap -sn 192.168.1.0/24

Output:

```

(kali㉿kali)-[~]
$ nmap -sn 192.168.1.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-23 11:12 EDT
Stats: 0:00:07 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 3.42% done; ETC: 11:15 (0:03:18 remaining)
Stats: 0:00:10 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 4.88% done; ETC: 11:15 (0:03:15 remaining)
Stats: 0:00:19 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 9.28% done; ETC: 11:15 (0:03:06 remaining)
Stats: 0:00:26 elapsed; 0 hosts completed (0 up), 256 undergoing Ping Scan
Ping Scan Timing: About 12.21% done; ETC: 11:15 (0:03:07 remaining)
Nmap scan report for 192.168.1.0
Host is up (0.00090s latency).
Nmap scan report for 192.168.1.1
Host is up (0.00088s latency).
Nmap scan report for 192.168.1.2
Host is up (0.00046s latency).
Nmap scan report for 192.168.1.3
Host is up (0.00050s latency).
Nmap scan report for 192.168.1.4
Host is up (0.0017s latency).
Nmap scan report for 192.168.1.5
Host is up (0.0011s latency).
Nmap scan report for 192.168.1.6
Host is up (0.00041s latency).
Nmap scan report for 192.168.1.7
Host is up (0.00045s latency).
Nmap scan report for 192.168.1.8
Host is up (0.0012s latency).
Nmap scan report for 192.168.1.9
Host is up (0.00045s latency).
Nmap scan report for 192.168.1.10
Host is up (0.00064s latency).
Nmap scan report for 192.168.1.11
Host is up (0.00061s latency).
Nmap scan report for 192.168.1.12

```

Result:

Successfully discovered active hosts.

Experiment 14: Password Cracking Using Hydra

Aim: To perform password auditing using Hydra.

Procedure:

- 1) Create password file (nano passwords.txt – opens editor)
- 2) Enter passwords (type inside editor)
- 3) Save file (Ctrl + O + Enter – saves content)
- 4) Exit editor (Ctrl + X – closes nano)
- 5) Run Hydra (hydra -l admin -P passwords.txt ftp://192.168.1.10 – performs password attack)

Commands:

- 1) nano passwords.txt
- 2) hydra -l admin -P passwords.txt <ftp://192.168.1.10>

Output:

```
(kali@kali)-[~]
└─$ nano passwords.txt

(kali@kali)-[~]
└─$ hydra -l admin -P passwords.txt ftp://127.0.0.1
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

hydra: invalid option -- '1'

(kali@kali)-[~]
└─$ hydra -l admin -P passwords.txt ftp://127.0.0.1
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-23 11:29:17
[DATA] max 5 tasks per 1 server, overall 5 tasks, 5 login tries (l:1/p:5), ~1 try per task
[DATA] attacking ftp://127.0.0.1:21/
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-07-23 11:29:20

(kali@kali)-[~]
└─$
```

Result:

Successfully demonstrated password auditing using Hydra.

Experiment 15: Information Gathering Using theHarvester

Aim: To gather publicly available information about a domain.

Procedure:

- 1) Run theHarvester (theHarvester -d example.com -b google – collects emails and subdomains)

Commands:

- 1) theHarvester -d example.com -b google

Output:

```
(kali㉿kali)-[~]
└─$ theHarvester -d google.com -b bing
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* | _ | _ |   ^   ^   | _ | _ |   *
* | _ | _ |   ^   ^   | _ | _ |   *
* | _ | _ |   ^   ^   | _ | _ |   *
* \_/_/_/_/_/_/_/_/_/_/_/_/_/_/_ *
*
* theHarvester 4.8.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: google.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
    Searching 0 results.
[*] Searching Bing.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] Hosts found: 0
```

Result:

Collected emails, subdomains, and publicly available information.

Experiment 16: Reconnaissance Using Google and Whois

Aim: To gather domain information.

Procedure:

1. Get domain info (whois example.com – displays domain registration details)
2. Search site (site:example.com – lists indexed pages)
3. Search files (filetype:pdf site:example.com – finds PDF documents)

Commands:

1. `whois example.com`
2. `site:example.com`

3. filetype:pdf site:example.com

Output:

```
(kali@kali)-[~]
$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-23T15:48:07Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.
```

Result:

Successfully gathered reconnaissance information.

Experiment 17: Windows OS Commands Execution

Aim: To study common Windows networking commands.

Procedure:

1. Trace route (tracert google.com – shows path packets travel)
2. Check connectivity (ping google.com – tests network reachability)
3. Show IP config (ipconfig – displays IP address)
4. Detailed config (ipconfig /all – full network details)
5. Show connections (netstat -an – lists active connections)

Commands:

1. traceroute google.com
2. ping google.com
3. ipconfig
4. ipconfig /all
5. netstat -an

Output:

```
(kali@kali)~$ ping google.com
PING google.com (142.250.206.78) 56(84) bytes of data:
64 bytes from pomaas-py-in-f14.1e380.net (142.250.206.78): icmp_seq=1 ttl=64 time=31.9 ms
64 bytes from pomaas-py-in-f14.1e380.net (142.250.206.78): icmp_seq=2 ttl=64 time=78.7 ms
64 bytes from pomaas-py-in-f14.1e380.net (142.250.206.78): icmp_seq=3 ttl=64 time=206 ms
64 bytes from pomaas-py-in-f14.1e380.net (142.250.206.78): icmp_seq=4 ttl=64 time=52.6 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 712ms
rtt min/avg/max/mdev = 31.798/85.286/163.785/45.627 ms

(kali@kali)~$ traceroute google.com
traceroute to google.com (142.250.206.78), 30 hops max, 60 byte packets
 1 10.0.2.2 (10.0.2.2)  5.829 ms  6.727 ms  4.393 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
```

```
(kali@kali)~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::66cb:297d:718a:d5c7 prefixlen 64 scopeid 0<link>
    inet6 fd17:625c:f037:2:c895:be2a:58f0:bb0e prefixlen 64 scopeid 0<global>
    ether 08:00:27:63:b0:05 txqueuelen 1000 (Ethernet)
    RX packets 5613 bytes 880521 (859.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 6446 bytes 423634 (413.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 10258 bytes 433406 (423.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10258 bytes 433406 (423.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)~$
```

Result:

Successfully executed Windows networking commands.